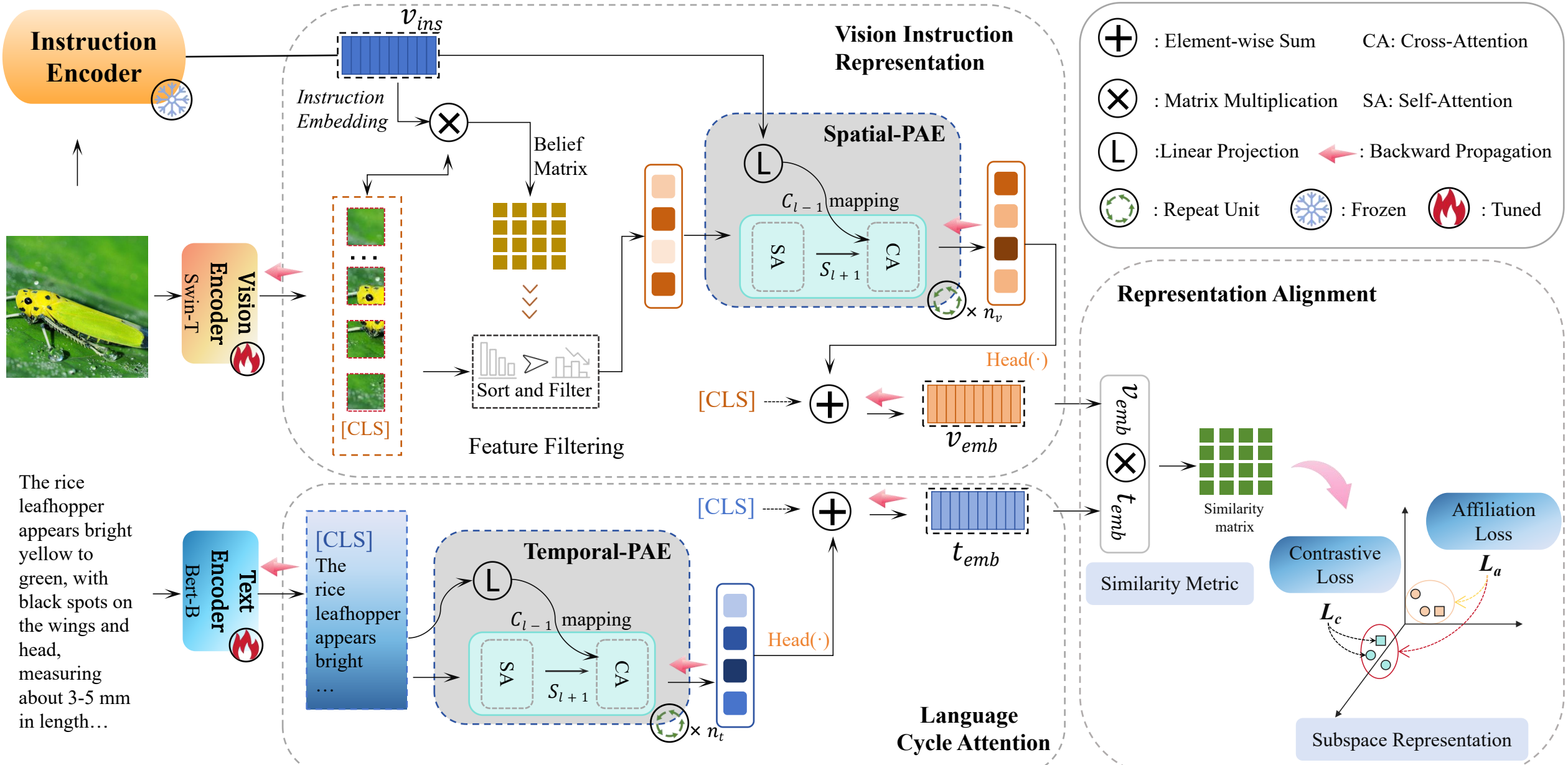
Instruction Encoder
v_{ins}
Instruction Embedding
Belief Matrix
Vision Instruction Representation
Spatial-PAE
C_{l-1} mapping
SA
S_{l+1}
CA
$\times n_v$
Vision Encoder
Swin-T
[CLS]
Sort and Filter
Feature Filtering
Head(·)
[CLS]
v_{emb}
: Element-wise Sum
CA: Cross-Attention
: Matrix Multiplication
SA: Self-Attention
:Linear Projection
: Backward Propagation
: Repeat Unit
: Frozen
: Tuned
Representation Alignment
The rice leafhopper appears bright yellow to green, with black spots on the wings and head, measuring about 3-5 mm in length…
Text Encoder
Bert-B
[CLS]
The rice leafhopper appears bright …
Temporal-PAE
C_{l-1} mapping
SA
S_{l+1}
CA
$\times n_t$
Head(·)
[CLS]
t_{emb}
Language Cycle Attention
v_{emb}
t_{emb}
Similarity matrix
Similarity Metric
Contrastive Loss
Affiliation Loss
L_a
L_c
Subspace Representation